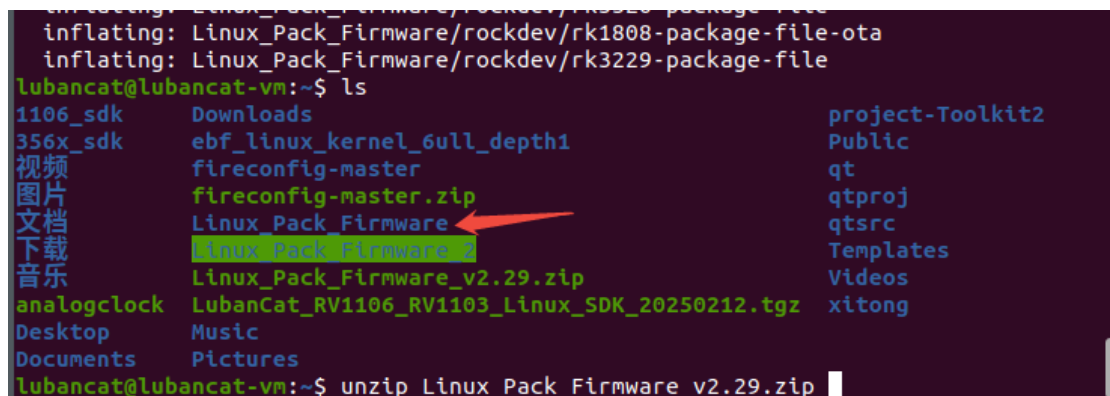


0. 环境准备

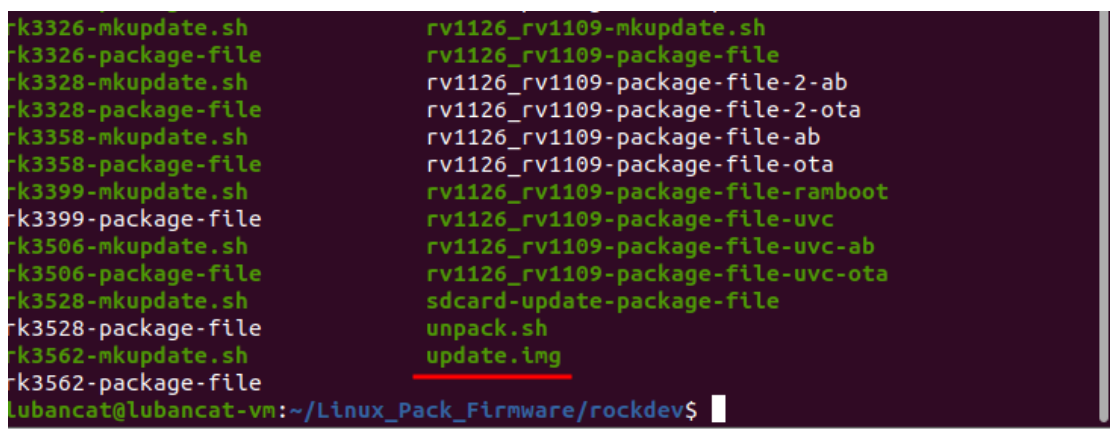
下载网盘 6-开发软件里的“Linux_Pack_Firmware_v2.29.zip”，这个是解包和打包的工具，同时准备好要解包的镜像，我这里用 “lubancat-rk3576-ubuntu22.04-gnome-20250721_update.img” 示例。

把解包和打包工具解压到虚拟机，如图：



```
inflating: Linux_Pack_Firmware/rockdev/rk1808-package-file-ota
inflating: Linux_Pack_Firmware/rockdev/rk3229-package-file
lubancat@lubancat-vm:~$ ls
1106_sdk      Downloads      project-Toolkit2
356x_sdk      ebf_linux_kernel_6ull_depth1  Public
视频          fireconfig-master  qt
图片          fireconfig-master.zip  qtproj
文档          Linux_Pack_Firmware  qtsrc
下载          Linux_Pack_Firmware_v2.29.zip  Templates
音乐          LubanCat_RV1106_RV1103_Linux_SDK_20250212.tgz  Videos
analogclock   Music          xitong
Desktop       Pictures
Documents
```

把要解包的镜像解压到./Linux_Pack_Firmware/rockdev 里，并且重命名为 update.img，为了不出现权限问题，可以执行 `sudo chmod 666 update.img` 如图：



```
rk3326-mkupdate.sh      rv1126_rv1109-mkupdate.sh
rk3326-package-file     rv1126_rv1109-package-file
rk3328-mkupdate.sh      rv1126_rv1109-package-file-2-ab
rk3328-package-file     rv1126_rv1109-package-file-2-ota
rk3358-mkupdate.sh      rv1126_rv1109-package-file-ab
rk3358-package-file     rv1126_rv1109-package-file-ota
rk3399-mkupdate.sh      rv1126_rv1109-package-file-ramboot
rk3399-package-file     rv1126_rv1109-package-file-uvc
rk3506-mkupdate.sh      rv1126_rv1109-package-file-uvc-ab
rk3506-package-file     rv1126_rv1109-package-file-uvc-ota
rk3528-mkupdate.sh      sdcard-update-package-file
rk3528-package-file     unpack.sh
rk3562-mkupdate.sh      update.img
rk3562-package-file
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev$
```

1. 解包

执行./unpack.sh，等待解压完成，解压完成后会有 output 文件夹存放解包产物，使用 `cp -r output/ Image/` 复制一份，把 Image 文件夹里的 package-file 复制到和打包脚本同一级目录下，如图：

```

boot.img MiniLoaderAll.bin package-file parameter.txt rootfs.img uboot.img
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev/Image$ cp package-file ../
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev/Image$ cd ..
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev$ tree
.
├── afptool
├── Image
│   ├── boot.img
│   ├── MiniLoaderAll.bin
│   ├── package-file
│   ├── parameter.txt
│   ├── rootfs.img
│   └── uboot.img
├── orion-mkupdate.sh
├── orion-package-file
├── output
│   ├── boot.img
│   ├── MiniLoaderAll.bin
│   ├── package-file
│   ├── parameter.txt
│   ├── rootfs.img
│   └── uboot.img
├── package-file
├── px30-mkupdate.sh
├── px30-package-file
├── px3se-mkupdate.sh
├── px3se-package-file
├── readme.txt
└── revision.txt

```

2. 替换

以替换 rootfs 为例子,我从外面复制要替换的 rootfs.img 到 Image/目录里,为了避免权限问题,我们还是确保 rootfs.img 的权限和 Image/目录里其他镜像一样,执行 `sudo chmod 664 rootfs.img` 修改权限, 如图:

```

lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev/Image$ ls -la
总用量 9495508
drwxrwxr-x 2 lubancat lubancat    4096 12月 19 08:52 .
drwxrwxr-x 4 lubancat lubancat    4096 12月 19 08:54 ..
-rw-rw-r-- 1 lubancat lubancat 134217728 12月 19 08:48 boot.img
-rw-rw-r-- 1 lubancat lubancat  481728 12月 19 08:48 MiniLoaderAll.bin
-rw-rw-r-- 1 lubancat lubancat    151 12月 19 08:48 package-file
-rw-rw-r-- 1 lubancat lubancat     348 12月 19 08:48 parameter.txt
-rw-rw-r-- 1 root     root  4718592000 12月 18 15:27 rootfs.img
-rw-rw-r-- 1 lubancat lubancat  8388608 12月 19 08:48 uboot.img
-rw-rw-r-- 1 lubancat lubancat 4861687812 12月 19 08:54 update.img

```

3. 打包

先修改打包的脚本,我这边用 rk3576 的镜像,所以修改 3576 的打包脚本。使用 `gedit rk3576-mkupdate.sh` 修改这部分, 如图:

```

8 echo "start to make update.img..."
9 if [ ! -f "Image/parameter" -a ! -f "Image/-
  parameter.txt" ]; then
10     echo "Error:No found parameter!"
11     exit 1
12 fi
13 if [ ! -f "package-file" ]; then
14     echo "Error:No found package-file!"
15     exit 1
16 fi
17 ./afptool -pack ./Image Image/update.img || pause
18 ./rkImageMaker -RK3576 Image/MiniLoaderAll.bin Image/
  update.img update.img -os_type:androidos || pause
19 echo "Making ./Image/update.img OK."

```

保存退出，然后删掉一开始解包的那个 update.img，再执行打包脚本，等待打包完成。打包完成后，打包脚本同级目录下会出现新的 update.img，如图：

```

rk3528-package-file
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev$ ./rk3576-mkupdate.sh
start to make update.img...
Android Firmware Package Tool v2.29
----- PACKAGE -----
Add file: ./Image/package-file
package-file,Add file: ./Image/package-file done,offset=0x800,size=0x97,userspace=0x1
Add file: ./Image/parameter.txt
parameter,Add file: ./Image/parameter.txt done,offset=0x1000,size=0x19b,userspace=0x1,fl
ash_address=0x00000000
Add file: ./Image/MiniLoaderAll.bin
bootloader,Add file: ./Image/MiniLoaderAll.bin done,offset=0x1800,size=0xbc1f9,userspace
=0x179
Add file: ./Image/uboot.img
uboot,Add file: ./Image/uboot.img done,offset=0xbe000,size=0x400000,userspace=0x800,flas
h_address=0x00004000
Add file: ./Image/boot.img
boot,Add file: ./Image/boot.img done,offset=0x4be000,size=0x8000000,userspace=0x10000,fl
ash_address=0x00008000
Add file: ./Image/rootfs.img
rootfs,Add file: ./Image/rootfs.img done,offset=0x84be000,size=0x100e00000,userspace=0x2
01c00,flash_address=0x00048000
Add CRC...
Make firmware OK!
----- OK -----
*****rkImageMaker ver 2.29*****
Generating new image, please wait...
Writing head info...
Writing boot file...
Writing firmware...
Generating MD5 data...
MD5 data generated successfully!
New image generated successfully!
Making ./Image/update.img OK.

```

```

rk3506-mkupdate.sh      rv1126_rv1109-package-file-uvic-ota
rk3506-package-file     sdcard-update-package-file
rk3528-mkupdate.sh      unpack.sh
rk3528-package-file     update.img
lubancat@lubancat-vm:~/Linux_Pack_Firmware/rockdev$

```

这个时候就可以拿去烧录了。